

Supplement: Experimental Details for Segment Running Time Estimation

I. EXPERIMENTAL SETTINGS

The primary experiment limits training and checkpoint-selection labels. This supplement reports additional development analyses of interval coverage, complete-label prefix objectives, training budgets, and total-duration sensitivity.

II. MATCHED OPTIMIZER-RECIPE CHECK

A separate matched comparison used the optimizer recipe selected from visible validation labels (cosine decay, initial learning rate 0.001, and 960 updates) and nine paired runs per budget. It evaluates the same Astana development trips as the main paper, but represents a distinct fitted-model configuration; it is not an independent dataset replication and does not replace the budget-specific main comparison. Ours-minus-direct MAE differences are +0.005, +0.133, and +0.148 seconds at 1%, 5%, and 10%, respectively. The 5,000-replicate trip-cluster percentile intervals condition on the fitted models and masks, so they quantify development-trip sampling uncertainty only and exclude retraining uncertainty. The interval at 1% includes zero; the intervals at 5% and 10% are positive. These conditional intervals do not establish generalization beyond this development split.

III. INDIVIDUAL LAYOUT CONTROLS

Both layouts expose 23,720 training labels across the same 115 intervals, preserving each interval’s count; validation exposes 2,656 labels. Training uses 240 updates, learning rate 0.003, and matched initialization and batches. Table II reports three repetitions. All matched-random Ours fits select the final update, linking the results to this training budget.

Count matching controls the number of labels per interval but does not balance their trips, departure periods, or contexts. It helps examine simultaneous changes in position and coverage without identifying their separate causal contributions.

IV. SINGAPORE PREFIX VERIFICATION

The comparison uses 50,000 training, 5,000 checkpoint-selection, and 10,000 development trips with complete local labels and no additional historical inputs. Training lasts 240 updates with selection every 24. Networks, initial states, batches, and duration weights are matched; the normalization denominator, prefix coefficient, or prediction form is changed. Counting padding includes near-zero cumulative errors after the trip ends and dilutes the objective. Valid-position normalization counts actual intervals only.

Table III gives means and sample SD, and Table IV gives individual repetitions. Valid-position normalization reduces overall MAE but increases long-interval MAE from 93.069

to 96.582 seconds; no-prefix fitting gives 97.566. Compared with no-prefix fitting, overall MAE decreases by 0.026, 0.056, and 0.055 seconds in the three pairs. Normalization affects the metrics differently.

V. SINGAPORE COMMON-VISIBLE COMPARISON

This known-total comparison uses 14,933 OD trips and 295,318 intervals and differs from both the Singapore prefix study and the Astana experiment. Common-visible methods use the same supplied OD total without same-segment history. The strict-past reference uses labels from strictly earlier trips and is reported separately because its information set differs. Ours uses the same bounded redistribution structure as the main method, with 15 input features, complete local labels, and duration-weighted Huber and prefix objectives. The comparison in the main paper uses a padding-inclusive prefix denominator; the valid-position variant is reported separately in the prefix experiment. It uses width 64, two Transformer layers, four heads, batch size 2,048, 240 updates, and learning rate 0.003; the prefix coefficient is 0.35. Ours and learned adapters report mean $\pm$ population SD over three seeds; deterministic references have one run. MixTTE, DutyTTE, and TriDGNet [1]–[3] are same-task adaptations rather than full reproductions, and their parameter counts and training budgets are not matched. The supplied total is derived from labels and may differ from the sum of local interval labels, so these errors measure allocation under a supplied total rather than recovery from an independently measured aggregate. This evaluation group is descriptive and is not an independent test.

VI. TRAINING-BUDGET CHECKS

The ProbETA adaptation uses author code [4], fitting Gaussian likelihoods on 256 visible singleton draws per update with Adam, learning rate 0.003, 10% training labels, and all 26,561 checkpoint-selection labels. Selection occurs every 240 updates. This differs from the primary trip batches, limited checkpoint-selection labels, and Huber objective, so results are reported separately.

Increasing the budget from 960 to 3,840 updates while preserving initialization and the first 960 draws reduces evaluation MAE from 36.403 to 32.876 seconds. Validation selects updates 3,600, 3,840, and 3,360 (Table V). One remains at the final update, showing that the shorter budget limited performance and that optimization adequacy still needs examination.

The 1,920-update check in Table VI uses 256 visible singleton draws per update and complete validation labels. Learning rates are 0.001 for Ours and identity-only methods and 0.003 for direct prediction. All selected checkpoints precede the last update, followed by increased validation error. Because

TABLE I

SEPARATE MATCHED OPTIMIZER-RECIPE CHECK ON THE ASTANA DEVELOPMENT SET. MAE IS IN SECONDS; RUN SUMMARIES ARE MEAN $\pm$ SAMPLE SD OVER NINE PAIRED RUNS. THE PERCENTILE INTERVALS RESAMPLE TRIPS AND ARE CONDITIONAL ON THE FITTED MODELS AND LABEL MASKS.

Budget	Ours MAE	Direct MAE	Difference	95% trip-bootstrap CI
1%	33.191 $\pm$ 0.278	33.186 $\pm$ 0.605	+0.005	[−0.077, 0.088]
5%	29.590 $\pm$ 0.219	29.457 $\pm$ 0.333	+0.133	[0.063, 0.202]
10%	28.629 $\pm$ 0.153	28.481 $\pm$ 0.140	+0.148	[0.074, 0.223]

TABLE II

INDIVIDUAL INITIALIZATION-REPEAT MAES (S) FOR THE IDENTITY-COUNT CONTROL. P: PREFIX; R: MATCHED RANDOM.

Repeat	P Ours	P direct	R Ours	R direct
1	50.102	49.889	49.462	51.407
2	49.193	50.548	50.762	51.966
3	51.069	50.979	50.968	50.749

TABLE III

SINGAPORE PREFIX-NORMALIZATION CHECK: FULL TRAINING/VALIDATION LABELS, THREE INITIALIZATIONS AND NO HISTORICAL INPUTS. ERRORS IN SECONDS. BOLD MARKS THE LOWEST ERROR WITHIN EACH COMPARABLE GROUP.

Method	MAE	Prefix50	Long MAE
Ours: padding-inclusive	19.518 $\pm$ 0.057	55.86	93.07
Ours: valid-only	19.445 $\pm$ 0.115	55.09	96.58
Ours: no prefix loss	19.490 $\pm$ 0.129	55.41	97.57
Direct prediction: valid-only	19.545 $\pm$ 0.096	55.56	96.81

TABLE IV

INDIVIDUAL INITIALIZATION-REPEAT SINGAPORE MAES (S).

Repeat	Padding-inclusive	Valid-only	No prefix	Direct prediction
1	19.513	19.341	19.367	19.441
2	19.577	19.569	19.624	19.632
3	19.465	19.424	19.480	19.562

sampling, validation labels, and learning rates differ, this check analyzes training behavior rather than extending the primary ranking.

VII. COMPLETE VALIDATION LABELS AND TOTAL SENSITIVITY

Tables VII and VIII use all 26,561 validation labels to examine placement and calendar/correction branches. Prefix placement reduces interval coverage, while moving blackouts retain more intervals. The primary experiment further matches interval-specific counts. Complete and limited validation incur different label costs, so their results are interpreted separately.

Total sensitivity keeps the parameters of eighteen 10% models fixed and substitutes $T' = (1 + \epsilon)T$, giving 126 evaluations across seven biases. At a negative 5% bias, Ours MAE is 28.199 seconds, below the error with an exact total, but predictions no longer match the true total. The triangle inequality gives $\sum_i |t'_i - t_i| \geq |\epsilon|T$, so a common relative bias implies WAPE at least $|\epsilon|$. This experiment tests uniform scale bias rather than random measurement error.

VIII. ADDITIONAL HISTORY AND EFFICIENCY

The history extension uses up to 20,000 training-partition trips completed before each query with matching intervals.

They can lie outside the 50,000 optimization queries and therefore provide additional labels. Table X holds history input fixed and compares pre-rescaling interval log-time coefficients of 0 and 0.15. This experiment uses padding-inclusive prefix normalization and is reported separately from the no-history auxiliary comparison.

Timing uses an RTX 4090 D and PyTorch 2.5.1+cu121, saved 10% primary checkpoints, a fixed 256-trip batch, float32, ten warmups, and 30 synchronized forward measurements. Table XI reports median and interquartile range. Loading, transfers, and fitting are excluded.

REFERENCES

- [1] W. Jiang, J. Han, R. Han, and H. Liu, “MixTTE: Multi-level mixture-of-experts for scalable and adaptive travel time estimation,” in *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2026.
- [2] X. Mao, Y. Lin, S. Guo, Y. Chen, X. Xian, H. Wen, Q. Xu, Y. Lin, and H. Wan, “DutyTTE: Deciphering uncertainty in origin-destination travel time estimation,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 12, 2025, pp. 12 390–12 398.
- [3] J. Zuo, Y. Yao, and Y. Zhang, “TriDGNNet: Triple feature encoder-based dual granularity graph learning network for enhanced travel time estimation,” *IEEE Transactions on Intelligent Transportation Systems*, 2026.

TABLE V
FIXED 3,840-UPDATE CHECK OF THE PROBETA ADAPTATION; ERRORS IN SECONDS.

Repeat	Selected step	Evaluation MAE	Final validation MAE
1	3600	32.552	31.108
2	3840	32.999	31.355
3	3360	33.078	32.785

TABLE VI
1,920-UPDATE TRAINING CHECK: ONE 10% TRAINING MASK AND FULL
VALIDATION LABELS. LEARNING RATES DIFFER BY METHOD; THIS TABLE
DOES NOT RANK METHODS. SELECTED STEPS FOLLOW REPEATS 1/2/3;
MAE IN SECONDS.

Method	MAE	Selected steps
Ours	28.893	1200/1200/1680
Direct prediction	28.620	1440/480/1680
Identity inputs	28.946	1440/960/1200

- [4] C. Xu, Q. Wang, and L. Sun, "Link representation learning for probabilistic travel time estimation," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 11, pp. 21 149–21 161, 2025. [Online]. Available: <https://doi.org/10.1109/TITS.2025.3590075>

TABLE VII

LABEL LAYOUTS WITH 23,720 TRAINING LABELS, FULL VALIDATION LABELS AND THREE INITIALIZATIONS. IDS: VISIBLE INTERVAL IDENTITIES; LATER MAE: FINAL APPROXIMATELY 80% OF EACH TRIP (S). BOLD MARKS THE LOWEST ERROR WITHIN EACH COMPARABLE GROUP.

Layout	Method	IDs	All MAE	Later MAE
Random	Ours	301	28.522 ± 0.216	27.776
Random	Direct prediction	301	28.353 ± 0.212	27.641
Prefix 20%	Ours	115	50.121 ± 0.938	55.375
Prefix 20%	Direct prediction	115	50.472 ± 0.549	55.818
Moving blackout	Ours	310	28.774 ± 0.116	28.348
Moving blackout	Direct prediction	310	28.810 ± 0.013	28.401

TABLE VIII

CALENDAR AND CORRECTION CONTROLS WITH ONE 10% TRAINING MASK, FULL VALIDATION LABELS AND THREE INITIALIZATIONS. ERRORS IN SECONDS. BOLD MARKS THE LOWEST ERROR WITHIN EACH COMPARABLE GROUP.

Variant	MAE	Prefix50
No correction	29.035 ± 0.175	145.17
No calendar	28.612 ± 0.160	150.74
Ours	28.522 ± 0.216	141.74
Shuffled calendar	29.082 ± 0.298	157.30

TABLE IX

TOTAL-BIAS SENSITIVITY SCENARIOS USING FROZEN 10% MODELS: NINE RUNS PER METHOD. MAE IN SECONDS AND WAPE IN PERCENT. BIAS SCENARIOS ARE NOT RANKED.

Bias	Ours MAE	Direct MAE	Ours WAPE	Direct WAPE
-10%	28.361	28.337	30.58	30.56
-5%	28.199	28.149	30.41	30.35
-1%	28.500	28.425	30.73	30.65
+0%	28.633	28.551	30.87	30.79
+1%	28.788	28.699	31.04	30.95
+5%	29.629	29.507	31.95	31.82
+10%	31.150	30.981	33.59	33.41

TABLE X

SEPARATE SINGAPORE HISTORY-INPUT CONTROLS WITH THE SAME STRICTLY-PAST SOURCES. SECONDS; MEAN $\pm$ SAMPLE SD OVER THREE INITIALIZATIONS. BOLD MARKS THE LOWEST ERROR WITHIN EACH COMPARABLE GROUP.

Method	MAE	Prefix50	Tail MAE
History + loss	19.304 ± 0.116	54.603 ± 0.436	92.217 ± 2.536
History only	19.324 ± 0.106	54.606 ± 0.429	93.736 ± 2.335

TABLE XI

SAME-DEVICE 256-TRIP FORWARD TIMING (MS), 30 REPEATS. BOLD MARKS THE LOWEST MEDIAN.

Method	Median	Interquartile range
Ours	6.040	[6.034, 6.043]
Direct prediction	5.450	[5.435, 5.469]